From Genes to Music

Introduction

The code of life is inherently rhythmic, structural, and profoundly multi-dimensional. While modern multi-omics technologies have successfully decoded the molecular inventories of cells—spanning genomic sequences, epigenomic landscapes, transcriptomic dynamics, and folded protein architectures—the sheer high-dimensionality of these datasets creates formidable barriers for human intuition and interpretation. For decades, biology has relied almost exclusively on visual representations (e.g., scatter plots, heatmaps, and network graphs) to decipher these complexities. However, visual paradigms can be less intuitive for conveying simultaneous temporal dynamics, layered streams, and hierarchical topologies, especially when users must compare multiple variables across time or scale. In response, the sonification and musification of biological data have emerged as a powerful orthogonal paradigm, transforming complex molecular information into auditory and musical forms.[1, 2] Beyond technical data display, this movement has also been shaped by collaborations between scientists, musicians, composers, and sound artists, reflecting the dual identity of sonification as both an analytical interface and a cultural practice of scientific communication.[3] At the same time, auditory display research emphasizes that effective sonification must be constrained by human perceptual capacities rather than by data availability alone.[4]

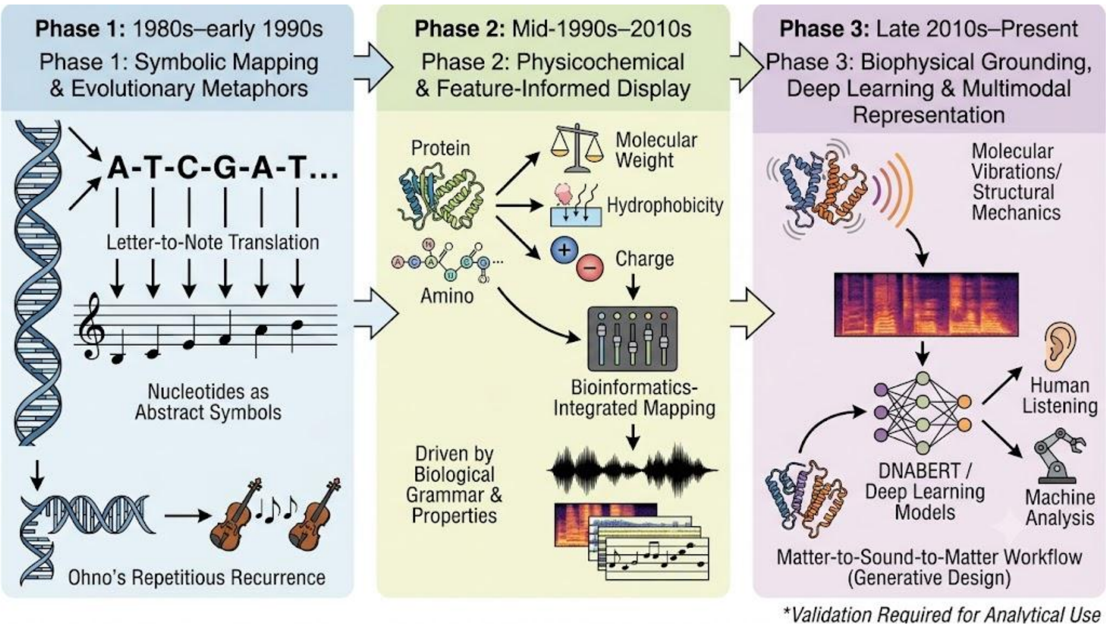
By mapping biological variables onto musical parameters such as pitch, rhythm, timbre, dynamics, and texture, researchers and artists can explore omics datasets through active listening. Because the human auditory system is highly sensitive to temporal patterns, frequency shifts, and layered acoustic streams, musification can reveal recurrent motifs, structural tensions, and dynamic coordinated behaviors that are often obscured in conventional visual displays. Consequently, this acoustic translation provides new avenues for exploratory data analysis, bioinformatics communication, accessibility for visually impaired researchers, and data-driven musical composition.[1, 2]

Despite a growing proliferation of methods, the field remains highly fragmented. Existing applications range from analytical sonification tools designed for rigorous sequence inspection to purely artistic compositions loosely inspired by DNA or protein datasets. These efforts differ fundamentally in their biological assumptions, mapping strategies, musical objectives, and intended audiences. Furthermore, many approaches rely on arbitrary "one-size-fits-all" algorithms that ignore the distinct physical and topological properties of different molecular layers. As a result, interdisciplinary practitioners often struggle to find a systematic overview of the field's conceptual foundations, methodological evolution, and scientific limitations.

To bridge this gap, this review provides a comprehensive, modality-specific framework for the musification of biomolecular and multi-omics data. We first trace the historical evolution of the field from early symbolic sequence-to-note translations to modern biophysically grounded and artificial-intelligence-assisted multimodal representations. Subsequently, we systematically dissect tailored acoustic mapping strategies for genomic, epigenomic, transcriptomic, proteomic, metabolomic, and association-study data. We highlight how the distinct biological syntax of each layer—from the discrete linear codes of DNA to dynamic RNA expression matrices, folded protein architectures, metabolic networks, and population-level association landscapes—dictates its corresponding musical expression.

By integrating perspectives from bioinformatics, structural biology, music technology, and auditory display, this review aims to provide a rigorous and structured entry point for life scientists, composers, and cross-disciplinary researchers. Ultimately, we demonstrate how biomolecular musification transcends its origins as an artistic curiosity, maturing into a promising complementary mode of biological representation that holds profound potential for both musical innovation and scientific discovery.

The Historical Evolution of Genetic and Biomolecular Musification



Tracing the evolution of biological data sonification provides essential context for modern mapping strategies. In this review, sonification refers broadly to the transformation of data into non-speech sound, whereas musification organizes data-derived sound through musical parameters such as pitch, rhythm, harmony, timbre, register, and form.[1] Musification can therefore be considered a music-oriented subset or extension of sonification, but aesthetic organization alone does not guarantee analytical validity.

Over the past four decades, the musification of genes, proteins, and cellular networks has progressed from simple symbolic translation to biologically informed mapping and, more recently, to multimodal representations that support human listening, artistic creation, and machine-learning-assisted analysis. At present, however, most musification systems should not be interpreted as enabling unaided listeners to infer biological function directly. Their value depends on mapping design, perceptual evaluation, and, when used analytically, validation against independent biological labels or tasks.

Box 1 | Key transitions in biomolecular musification

Historical transition	Approximate period	Representative examples	Significance
From sequence letters to musical notes	**1980s**	**Hayashi & Munakata; Ohno & Ohno; Riego et al.**	**Established DNA musification as symbolic translation and conceptual analogy.**
From symbols to molecular features	**1990s–2000s**	**Gena & Strom; King & Colin; Dunn & Clark; Takahashi & Miller**	**Introduced physicochemical and sequence-derived properties into musical mapping.**
From ad hoc mapping to perceptual design	**2000s**	**Barrass**	**Emphasized that auditory displays should be designed around perceptual discriminability, task relevance, and listener interpretation rather than arbitrary parameter assignment.**
From isolated sequences to bioinformatics contexts	**2010s**	**Temple; Martin et al.**	**Incorporated reading frames, codons, protein sequences, and multiple sequence alignments into auditory display.**
From linear sequences to omics-scale data	**2010s**	**Staege; Cittaro et al.; Lukoševičius et al.**	**Expanded musification to gene expression, epigenomic tracks, and multilayered biological datasets.**
From artistic display to biophysical grounding	**2010s**	**Wong et al.; Yu et al.**	**Linked musical mapping to protein architecture, molecular vibrations, and structural mechanics.**
From sound output to multimodal	**2020s**	**Slenter; Wang et al.; Goutham & Narayanan**	**Positioned sound and music as potential intermediate representations for systems biology and**

Historical transition	Approximate period	Representative examples	Significance
representation			machine-learning-assisted analysis.

Phase 1: Symbolic Mapping and Evolutionary Metaphors (1980s to early 1990s)

Early efforts in biological musification treated DNA primarily as a symbolic sequence, equating nucleotides with text characters or musical notes. Hayashi and Munakata pioneered DNA-to-music conversion by mapping the four nucleotide bases to specific musical pitches, demonstrating that biological sequences could be rendered as auditory patterns.[5] Despite its simplicity, this work established a crucial foundation: biological sequences could be experienced as ordered temporal structures rather than merely read as alphabetical strings.

Subsequently, Susumu Ohno proposed the principle of "repetitious recurrence," drawing an analogy between repeated genetic motifs and thematic recurrence in classical music.[6] By converting gene sequences, such as the human phosphoglycerate kinase gene, into violin melodies, Ohno framed DNA as a temporally organized symbolic system. From a modern methodological standpoint, however, these works are best understood as conceptual and artistic cornerstones rather than validated analytical tools. Because such mappings depend heavily on choices of scale, tuning, tempo, and orchestration, they should be interpreted as symbolic translations rather than direct acoustic manifestations of DNA. Related early work on the "sound of the DNA language" further reinforced the idea that nucleotide sequences could be treated as ordered acoustic symbols, extending the DNA-to-sound metaphor beyond isolated artistic demonstrations and contributing to the broader conceptual foundation of genomic sonification.[7]

Phase 2: Physicochemical Mapping and Feature-Informed Auditory Display

By the mid-1990s, researchers increasingly treated nucleotides and amino acids as molecular entities with measurable physicochemical properties—such as molecular weight, hydrophobicity, charge, and structural preferences—rather than as abstract symbols alone. This paradigm shift propelled the field beyond literal "letter-to-note" translation toward feature-informed mapping.

In an early electronic-art context, Gena and Strom advanced DNA musical synthesis by integrating biologically meaningful parameters such as base composition and hydrogen-bonding capacity.[8] Dunn and Clark subsequently popularized protein musification in Life Music, linking amino acid properties to musical register and texture.[9] Around the same period, King and Colin introduced PM, a protein-music program that further illustrates the emergence of computational tools for translating amino-acid sequences into musical material.[10] Takahashi and Miller later refined amino acid sonification by grouping chemically similar residues and incorporating codon-level information.[11] Together, these works illustrate a shift from arbitrary symbolic mapping toward biologically motivated musical organization.

Concurrently, sonification became more closely integrated with bioinformatics. Temple developed a DNA sonification framework that incorporated reading-frame logic and start/stop codon structure, while Martin and colleagues extended auditory display techniques to protein multiple sequence alignments.[12, 13] These studies marked a shift from isolated symbol-to-note translation toward mappings informed by biological grammar and comparative sequence analysis.

This era also pushed sonification beyond linear DNA and protein sequences into more complex biological data types, including epigenetic signals, gene expression profiles, microbial sequence data, and doctoral work on metabolic network transformations.[14-18] These efforts broadened the field from sequence-based translation toward auditory representations of quantitative tracks and multi-layered biological relationships. The key lesson from this phase is that meaningful sonification requires biological features to guide musical mapping, rather than relying on arbitrary symbol-to-note assignment.

Phase 3: Biophysical Grounding, Deep Learning, and Multimodal Representation (Late 2010s to Present)

Since the late 2010s, biomolecular sonification has increasingly intersected with molecular biophysics, artificial intelligence, and multimodal representation learning, building on earlier protein-music analogies and materials-by-design work from the early 2010s.[19] Advancing this concept, Yu and colleagues developed a self-consistent sonification method for amino acid sequences.[20] The same matter–music framework was later extended conceptually to the SARS-CoV-2 spike protein, where musical and mechanical analogies were used to discuss viral infection, spike architecture, and biomolecular mechanics.[21] Rather than assigning notes arbitrarily, they calculated intrinsic amino acid vibrational frequencies and transposed them into the human auditory range. They further connected these sonified representations with recurrent neural networks for de novo protein sequence generation.[20]

This study expanded biomolecular sonification from auditory display toward a potential intermediate representation for computational design, suggesting a “matter-to-sound-to-matter” workflow. However, such workflows should be interpreted cautiously: they complement, rather than replace, molecular simulation, structure prediction, biochemical characterization, and experimental validation.

In parallel, biological language models such as DNABERT, together with studies on cross-domain transfer in pre-trained models, reinforced the idea that biological and musical materials can both be modeled as structured token sequences with learnable statistical patterns.[22, 23] Although not direct sonification studies, these works helped situate biomolecular musification within a broader landscape of sequence modeling and multimodal representation learning.

Recent work has also explored audio-derived representations, such as spectrograms and MFCCs, as machine-learning-ready feature spaces for protein function prediction.[24] In this context, sound becomes not only an artistic output for human listeners but also a possible intermediate representation for multimodal learning. Recent studies on NMR-informed protein sonification,[25] gene-expression musification,[26] and metabolic network transformations[18] further illustrate the field's expansion beyond simple linear sequences toward systems-level biological data.

Despite these developments, several methodological challenges remain unresolved. Many mappings remain underdetermined: the same biological feature can be assigned to multiple musical parameters, producing different perceptual interpretations. Musical coherence and biological fidelity may also conflict, and relatively few systems have been evaluated through controlled listening studies, benchmark machine-learning comparisons, or independent biological validation. Future work should therefore report mapping rationales, parameter choices, source code, datasets, and evaluation protocols more transparently.

Orchestrating the Omics: Modality-Specific Mapping Strategies

As the historical trajectory illustrates, the musification of biological data can no longer rely on a universal, one-size-fits-all algorithm. The structural properties of life's molecules vary drastically: genomic DNA presents as a discrete linear code, epigenetics as dynamic regulatory tracks, proteins as physically folded three-dimensional topologies, and metabolic pathways as complex interacting networks. Consequently, translating these distinct biological layers into sound demands tailored mapping frameworks, because auditory display is most informative when the acoustic variables are matched to the perceptual and structural properties of the underlying molecular information.[2] In the following sections, we systematically dissect the specific sonification strategies developed for each major biological data modality. We explore how the unique mathematical and biophysical features of genomic, transcriptomic, epigenomic, proteomic, and metabolomic data inform the selection of musical parameters, thereby transforming diverse molecular landscapes into coherent auditory experiences.

Genomic Data: From Sequence-to-Melody Translation to Analytical Sonification

The discrete linear code of genomes.

Genomic data are among the earliest and most widely explored materials in biological musification. Early DNA sonification studies, including Hayashi and Munakata's nucleotide-to-pitch mapping, Ohno and Ohno's recurrence-based musical analogy, and Riego and colleagues' exploration of the sound of DNA language, collectively established the genome as an ordered symbolic substrate for acoustic translation.[5-7] Unlike transcriptomic, proteomic, or metabolomic datasets, which are often numerical and condition-dependent, genomic sequences are linear, symbolic, and discrete. This makes them naturally suitable for rule-based mappings into musical parameters such as pitch, rhythm, register, instrumentation, and silence. However, existing approaches differ substantially in purpose. Some treat DNA as latent musical material to be shaped into aesthetically coherent compositions, whereas others treat sound as an auditory display for preserving and inspecting sequence information. The contrast between Susumu Ohno's music-oriented transformations and Mark D. Temple's analytical DNA sonification system illustrates this methodological spectrum particularly well.[6, 12]

Music-oriented transformation and repetitious recurrence.

Ohno's approach is grounded in the idea that genome evolution and musical composition share a principle of repetitious recurrence.[6] In this framework, coding sequences are not treated as random nucleotide strings, but as products of duplication, variation, and repeated use of primordial sequence modules. Ohno and Midori Ohno assigned nucleotide bases to pitch positions within an octave and then identified recurrent oligonucleotides as musical subjects. For example, in their transformation of the human X-linked phosphoglycerate kinase coding sequence, recurring sequence units were interpreted as principal or secondary themes and shaped into a violin score in D minor and 9/8 time.[6] The strength of this method lies in its musicality: repetition, motif, key, meter, and variation allow DNA-derived material to become aesthetically coherent. Its limitation, however, is that the final composition depends strongly on subjective musical judgment, including motif selection, tonal framing, and rhythmic organization. As a result, it is more valuable as a model for biologically inspired composition than as a rigorous tool for sequence analysis.

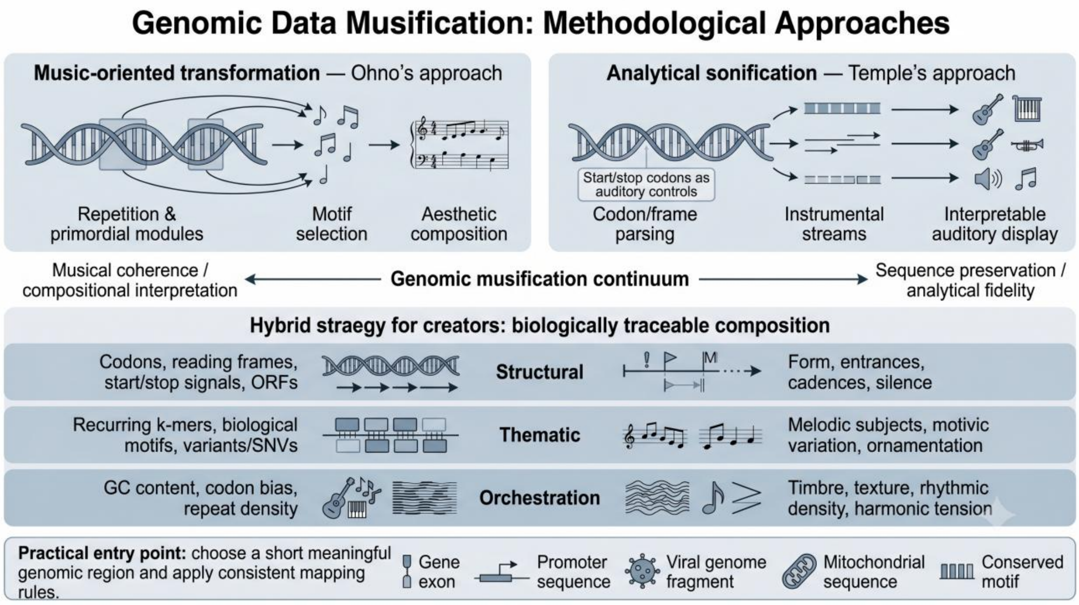
Analytical sonification and sequence fidelity.

By contrast, Temple's DNA sonification system emphasizes analytical fidelity.[12] Rather than transforming DNA into conventional music, Temple developed a browser-like auditory display in which nucleotide, dinucleotide, trinucleotide, codon, and reading-frame information can be mapped to sound. The most informative version parses codons simultaneously in all three possible reading frames, assigning each frame to a distinct instrumental stream. Start and stop codons can function as logical controls: start codons activate a stream, whereas stop codons silence it or are marked with percussive events.[12] This makes open reading frames, premature stops, repetitive sequences, and frame-shift-like disruptions audible as changes in continuity, silence, rhythm, or instrumentation. The advantage of this method is its transparent connection to biological sequence structure; its drawback is that the resulting sound may be repetitive, sparse, or aesthetically limited.

Together, these two approaches suggest that genomic musification exists on a continuum between artistic composition and analytical sonification. Ohno's work demonstrates how conserved motifs and sequence recurrence can generate musical themes, while Temple's work shows how codons, reading frames, and start-stop logic can be preserved in an interpretable auditory display. For creators who wish to use genomic data as musical inspiration without abandoning biological traceability, a useful entry point is therefore a hybrid strategy: use the biological structure of the sequence as the compositional skeleton, and use musical parameters to shape expression.

A hybrid strategy for biological traceability.
In practice, such a strategy can be organized across three layers. First, at the structural layer, the sequence may be parsed by codons or reading frames rather than by isolated nucleotides. Start codons, stop codons, open reading frames, repetitive elements, and conserved motifs can serve as formal markers, entrances, cadences, or points of silence. Second, at the thematic layer, frequently recurring k-mers or biologically meaningful motifs can be treated as melodic subjects, while mutations, single-nucleotide variants, or synonymous codons can generate variations in articulation, register, timbre, or ornamentation. Third, at the orchestration layer, different reading frames may be assigned to different instrumental groups, and sequence-level properties such as GC content, codon usage bias, or repeat density may control texture, brightness, rhythmic density, or harmonic tension. This approach avoids a purely arbitrary base-to-note translation while still allowing the genome to become musically expressive.

For artists and composers, the most accessible starting point is not to sonify an entire genome, but to choose a short biologically meaningful region, such as a gene exon, promoter sequence, viral genome fragment, mitochondrial sequence, or conserved motif. The creator can then define a small number of transparent mapping rules and keep them consistent throughout the piece. In this way, genomic musification can function both as a source of musical material and as a biologically grounded compositional method: the resulting work does not merely sound "inspired by DNA," but remains traceable to specific sequence features. A terminological caveat is also important. Not every computational method with a music-related name is a musification method. For example, MuSiCal is a recent framework for accurate and sensitive mutational signature analysis in cancer genomics, but its purpose is statistical decomposition of somatic mutation patterns rather than auditory or musical representation.[27] This distinction is useful because genomic musification should be defined by an explicit mapping from biological variables to sound, not by musical terminology alone.



Epigenomic Data: From Chromatin Tracks to Regulatory Music

Regulatory landscapes over fixed sequences.

Compared with genomic sequences, epigenomic data provide a more dynamic and quantitative substrate for biological musification. Rather than consisting of fixed symbolic characters, epigenomic profiles such as histone-modification ChIP-seq and DNA methylation describe regulatory states distributed along genomic coordinates. ChIP-seq data are typically represented as continuous enrichment signals or peaks, whereas DNA methylation can be modeled either as binary methylated/unmethylated states at individual CpG sites or as continuous beta values in bulk samples. These features make epigenomic data particularly suitable for mapping to musical parameters such as pitch, volume, duration, timbre, and polyphonic texture.

Multi-track chromatin music.

A representative example is the work of Cittaro et al., who proposed a proof-of-concept framework for converting histone-modification ChIP-seq signals into MIDI-based music. In their approach, genomic regions were divided into fixed-size bins, the average ChIP-seq intensity in each bin was log-transformed and discretized, and the resulting values were mapped to musical notes. Low-signal regions could be represented as rests, while stronger enrichment peaks generated higher or more salient musical events. Multiple chromatin marks, including H3K27me3, H3K27ac, H3K36me3, H3K4me3, H3K9me3 and Pol II, could also be combined into multi-track musical representations, suggesting the possibility of transforming combinatorial chromatin states into “chromatin chords.” Importantly, their study found that music generated from real ChIP-seq profiles was perceived as more appealing than music generated from randomized signals, although its ability to predict differential gene expression remained limited.[14]

Binary states and DNA methylation.

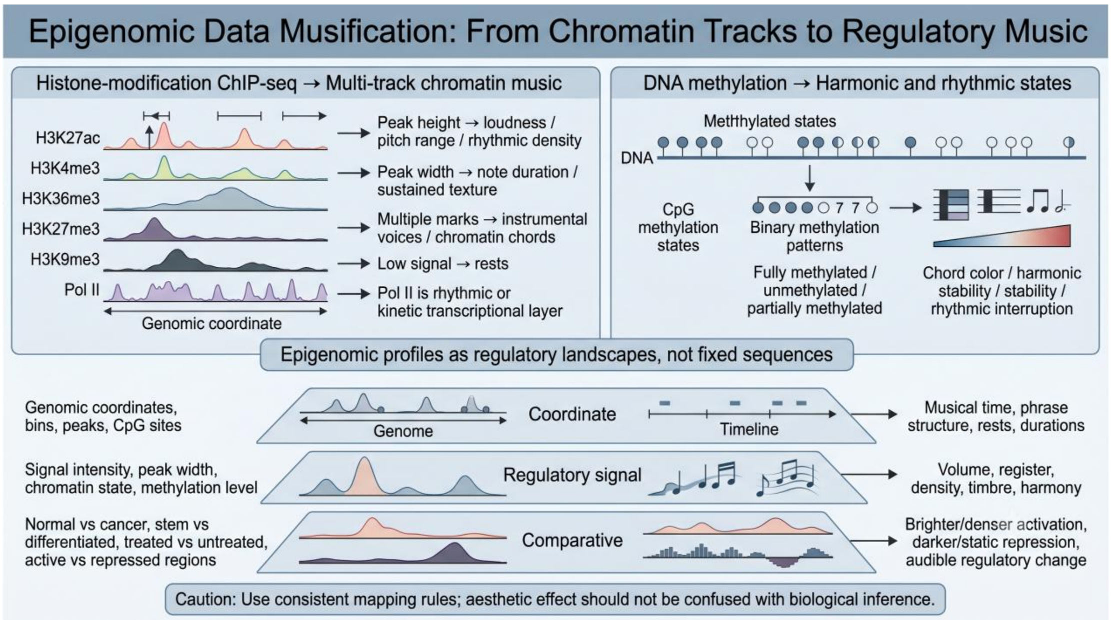
DNA methylation offers another form of epigenomic musification because methylation at individual CpG sites can be treated as a binary molecular state. Brocks developed a comparative epigenomics approach in which seven consecutive CpG sites were encoded as binary strings, generating 128 possible methylation patterns. These patterns were then mapped to a predefined universe of chords and rhythmic events. Fully methylated, unmethylated, and partially methylated fragments produced different musical structures, allowing methylation differences between normal, premalignant, and cancer samples to be represented audibly.[15] This approach illustrates how local methylation patterns and disease-associated epigenetic alterations can be transformed into harmonic and rhythmic variation.

Overall, epigenomic data have strong potential for biological music generation because they are dynamic, quantitative, and closely associated with gene regulation. For creators, however, epigenomic musification should be approached differently from genomic sequence musification. Rather than treating epigenomic data as a sequence-to-melody problem, it is more productive to treat them as regulatory landscapes that can be translated into musical texture, density, timbre, and form.

A practical entry point is to map genomic coordinates onto musical time and regulatory signals onto expressive parameters. ChIP-seq peak height can control loudness, pitch range, rhythmic density, or envelope intensity; peak width can influence note duration or sustained texture; and different histone marks can be assigned to different instrumental voices. For instance, active marks such as H3K27ac or H3K4me3 may be represented by brighter timbres, higher registers, or more rhythmically active material, whereas repressive marks such as H3K27me3 or H3K9me3 may be represented by darker timbres, lower registers, drones, or slower harmonic motion. Pol II occupancy can be used as a rhythmic or kinetic layer indicating transcriptional activity. DNA methylation can be translated into harmonic stability, rhythmic interruption, or chordal color, with fully methylated, unmethylated, and partially methylated regions producing distinct musical states.

A comparative paradigm for creators.

For artists and composers, the most effective use of epigenomic data may be comparative rather than absolute. Mapping absolute methylation or chromatin signals often risks producing static or monotonous textures. By applying identical mapping rules to contrasting states—such as normal versus cancer samples, stem-like versus differentiated cells, or treated versus untreated conditions—the music shifts from a static display to a narrative of regulatory transition. Activation may become brighter, denser, or more rhythmically mobile, whereas repression or hypermethylation may become darker, quieter, slower, or more fragmented. Nevertheless, aesthetic effectiveness should not be confused with biological inference; any interpretive claim should remain linked to the underlying data and, where possible, compared with conventional epigenomic analysis.



Transcriptomic Data: From Dynamic Expression Matrices to Gene Network Music

The high-dimensional canvas of gene expression.

Compared with genomic and epigenomic data, transcriptomic data introduce a more explicitly dynamic and high-dimensional substrate for biological musification. Because raw expression counts require normalization and often dimensionality reduction before interpretation, the musical mapping of transcriptomic data should generally be applied to processed and biologically meaningful representations rather than to raw count matrices. Transcriptomic profiles are not fixed symbolic sequences or static regulatory tracks, but matrices of gene expression values measured across time points, cell types, tissues, perturbations, or spatial coordinates. In bulk RNA-seq and microarray studies, transcriptomic data are commonly represented as expression matrices in which rows correspond to genes and columns correspond to samples or time points. In time-series transcriptomics, the same gene may display continuous temporal trajectories, delayed responses, oscillatory behavior, transient pulses, or sustained activation. In single-cell RNA-seq, the data further acquire a high-dimensional population structure in which individual cells occupy positions in a latent expression space, often organized into clusters, trajectories, or pseudotime continua. In spatial transcriptomics, expression values are additionally anchored to physical tissue coordinates, allowing gene activity to be interpreted not only as a molecular signal but also as a spatially distributed pattern of cellular organization.

Cellular polyphony and musical motion.
These properties make transcriptomic data especially suitable for music generation because they already possess features analogous to musical organization: temporal progression, variation, repetition, density, polyphony, and coordinated ensemble behavior. Whereas DNA sequence musification often maps discrete symbolic units to pitch and epigenomic musification maps regulatory landscapes to musical texture, transcriptomic musification can be understood as the translation of dynamic biological activity into musical motion. Gene expression values can be mapped to pitch, loudness, register, rhythmic density, or timbral intensity; time points can be mapped to beats, measures, phrases, or larger musical sections; and gene clusters can be represented as instrumental voices, contrapuntal lines, or harmonic layers.

Quantitative mapping and melodic generation.
Early work on gene expression musification treated expression intensity as a quantitative input for melodic generation. Staege proposed the Gene Expression Music Algorithm, or GEMusicA, as a method for transforming microarray-derived gene expression values into musical material. In this approach, expression signal intensities were converted into musical frequencies and tone durations, allowing different biological samples to produce distinct melodic profiles. GEMusicA was applied to gene expression datasets from cancer and stem-cell-related contexts, including neuroblastoma-derived cell lines, Ewing sarcoma-related analyses, Hodgkin lymphoma cell lines, and normal B cells. The algorithm was intended not only as an artistic translation of molecular data, but also as an alternative way to perceive differences in expression profiles between biological samples. Staege further argued that musical representations might help users detect sample-specific or disease-associated expression patterns, although such auditory interpretation should complement rather than replace conventional statistical analysis.[16]

Dimensionality reduction and harmonic intervals.
A related direction was developed by Alterovitz and Yuditskaya, who explored the sonification of high-dimensional gene expression and protein abundance dynamics. Their approach used dimensionality reduction, particularly principal component analysis, to transform complex biological measurements into musical structures. By mapping principal components of temporal molecular data to musical intervals and tuning relationships, they proposed that normal or control samples could generate more harmonious patterns, whereas abnormal or disease-associated profiles might produce more dissonant musical outputs. This comparative strategy is important because it shifts transcriptomic musification away from a single-gene melody model and toward an auditory representation of high-dimensional biological state.[28]

Layered frameworks and functional context.
More recently, Goutham and Narayanan proposed Gene Symphony, a two-layered musification framework for time-series gene expression clusters with functional backgrounds. Instead of musifying an entire gene set as a single global profile, they first clustered genes according to similar temporal expression patterns. This clustering step is biologically meaningful because co-expressed genes may reflect shared regulation, related biological function, or coordinated participation in cellular processes. Their foreground layer encoded expression dynamics across time points through pitch variation, so that changes in gene expression could be heard as melodic or chordal movement. Their background layer incorporated functional context by using Gene Ontology enrichment and semantic similarity scores, which were translated into smoothed audio modulated by intensity and echo effects. In this way, the listener could hear not only temporal expression trends but also the functional coherence of each gene cluster.[26]

The two-layered design of Gene Symphony is especially relevant for transcriptomic musification because it addresses a central limitation of simple expression-to-note mapping: expression values alone do not convey biological meaning unless they are interpreted within regulatory, functional, or network contexts. A single gene's increase or decrease may be musically audible, but its functional role often depends on its relationship to other genes. By combining a foreground melodic layer for expression dynamics with a background layer for GO-derived semantic cohesion, Goutham and Narayanan transformed the transcriptome into a layered musical object. In their reported user survey, participants were able to match audio clips to expression plots with an overall accuracy of 72.5%, and 88.9% perceived differences in background layers associated with semantic information. These preliminary results suggest that layered musification can communicate both temporal and functional aspects of expression data, though the study remains a preprint and its biological interpretability should be treated cautiously until further validation.[26]

Time-series trajectories and ensemble orchestration.
Building on expression-intensity mapping, dimensionality-reduction-based sonification, and layered gene-cluster musification, transcriptomic data invite a more explicitly polyphonic approach than many other biological data types.[16, 26, 28] Because thousands of genes are measured simultaneously, the transcriptome resembles an ensemble rather than a solo line. Gene clusters can therefore be mapped to instrumental sections, with each cluster contributing a voice to a larger musical texture. Genes with similar temporal activation may be assigned to one melodic line, while delayed or inverse expression profiles may be assigned to contrasting contrapuntal material. Upregulated clusters may be represented by increasing pitch, rising dynamics, or denser rhythmic activity, whereas downregulated clusters may be represented by falling contours, decreasing loudness, thinning texture, or lower registers. Co-expression modules may form harmonic blocks or rhythmic ostinati, while hub genes may be emphasized as leading melodic voices or recurring motifs.

Time-series gene expression is particularly well suited to musical temporal mapping because biological sampling points can be aligned directly with musical time. Early response genes may appear as short, high-energy motifs near the beginning of a piece; sustained expression programs may become drones, pedal tones, or long harmonic regions; and oscillatory genes may generate repeating rhythmic or melodic cycles. Developmental trajectories, circadian expression, immune activation, differentiation, or drug-response time courses can therefore be translated into musical forms that unfold as biological processes unfold. In single-cell contexts, pseudotemporal ordering further provides a way to arrange cells along inferred developmental or differentiation continua, making trajectory-based musical organization conceptually plausible.[29] In this sense, transcriptomic musification is not merely a translation of numerical values into notes, but a way of representing cellular change as temporal form.[26, 28, 29]

Prospective extensions to single-cell and spatial transcriptomics.

Single-cell transcriptomics expands this possibility by adding cellular heterogeneity. Instead of averaging expression across a population, scRNA-seq data represent large numbers of individual cells, each with its own expression profile, and are commonly interpreted through preprocessing, normalization, clustering, latent-space embedding, and trajectory or pseudotime analysis.[29-31] These features suggest possible musification strategies based on cell clusters, differentiation trajectories, pseudotime, lineage branching, or cell-state transitions. Distinct cell types may be assigned different timbres or instrumental families; pseudotime could be mapped to musical time; and transitions from stem-like to differentiated states could be represented as gradual changes in harmony, register, texture, or orchestration. Rare cell states or transitional populations could be made musically salient through distinctive motifs, unusual timbres, or spatialized sound. However, because scRNA-seq matrices are sparse and affected by technical noise, dropout, cell-cycle effects, and batch variation, musification strategies should be based on robust cell-state or trajectory-level features rather than on isolated single-gene fluctuations.[30, 31]

Spatial transcriptomics adds another potential layer because gene expression can be linked to anatomical position, allowing molecular measurements to be interpreted in relation to tissue architecture.[32, 33] Tissue coordinates could be mapped to stereo or surround-sound placement, so that spatial gene-expression patterns become spatialized musical events. Regions of high expression may generate dense or bright sonic clusters, whereas low-expression regions may become silence, sparse texture, or low-intensity ambience. Spatial gradients could be translated into pitch glissandi, timbral morphing, or sound movement across speakers. Spatial transcriptomics may therefore provide a promising substrate for representing tissue architecture, tumor microenvironments, developmental patterning, or cell-cell neighborhood effects. However, because spatial transcriptomic datasets are technically complex and often affected by segmentation, capture efficiency, resolution, and spot- or cell-level constraints, any artistic mapping should remain traceable to the original spatial data and avoid overclaiming biological interpretation.[32, 33] Tissue coordinates could be mapped to stereo or surround-sound placement, so that spatial gene-expression patterns become spatialized musical events. Regions of high expression may generate dense or bright sonic clusters, whereas low-expression regions may become silence, sparse texture, or low-intensity ambience. Spatial gradients could be translated into pitch glissandi, timbral morphing, or sound movement across speakers. Spatial transcriptomics may therefore provide a promising substrate for representing tissue architecture, tumor microenvironments, developmental patterning, or cell-cell neighborhood effects.

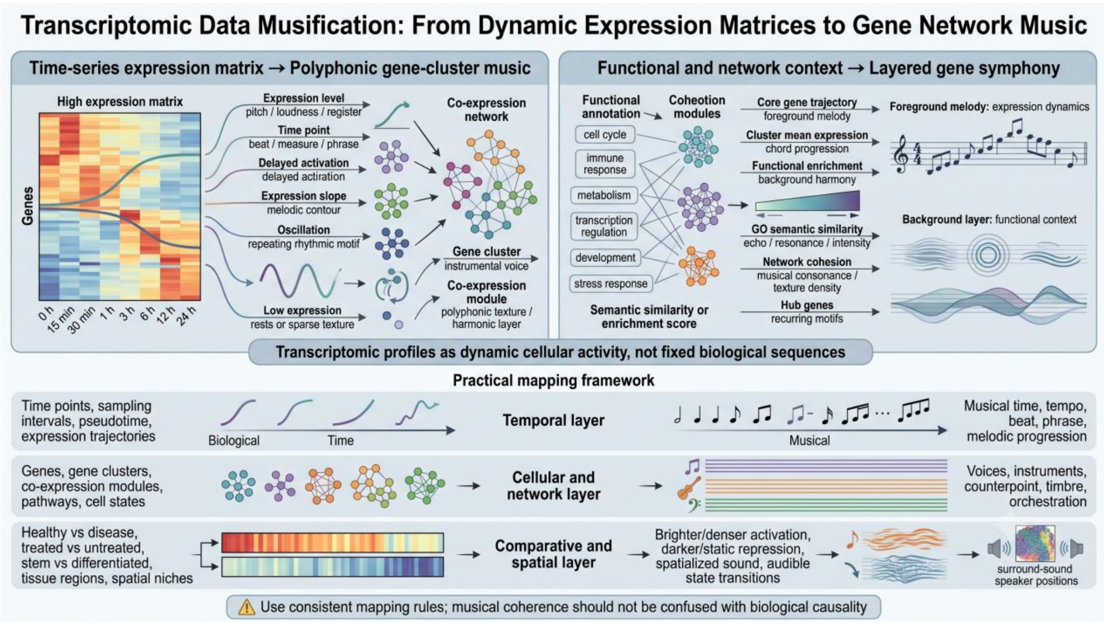
Design principle: cellular orchestration rather than gene-to-note translation.
Synthesizing previous expression-mapping, dimensionality-reduction, and layered cluster-based approaches, transcriptomic musification may be most powerful when treated as dynamic systems music rather than as a simple table-to-melody conversion.[16, 26, 28] A practical design begins by deciding what biological structure should control the musical structure. Time points may define musical chronology; gene clusters or functional pathways may define voices; expression levels may define pitch, loudness, density, or timbre; and functional annotations may define harmonic color or background texture. If the goal is comparison, the same mapping rules should be applied to multiple conditions, such as healthy versus diseased tissue, treated versus untreated cells, early versus late developmental stages, or one cell lineage versus another. In this comparative format, the audience hears not just expression values but biological change: activation becomes brighter, denser, more rhythmically active, or more harmonically resolved; repression becomes quieter, darker, slower, or more fragmented; and cell-state transition becomes audible as gradual musical transformation.

This orchestration strategy also addresses the central scaling dilemma of transcriptomic data: translating a single gene’s expression into melody strips away network context, while simultaneously sounding thousands of genes creates an unintelligible cacophony. The optimal compromise is to map co-expressed gene clusters, functional pathways, or cell states to distinct musical roles. For example, a metabolic pathway may be represented as a string section, whereas an immune-response module may be represented as brass or percussion. Highly variable hub genes may become foreground motifs, while stable regulatory networks may provide background harmonic textures. Conceptually, transcriptomic musification can therefore be framed as a form of cellular polyphony, in which genes, clusters, pathways, cell states, and spatial niches are assigned distinct but coordinated musical roles within a changing biological score.[16, 26, 28]

Methodological caveats and analytical limitations.
Nevertheless, transcriptomic musification faces important limitations. Gene-expression datasets are high-dimensional, noisy, and sensitive to normalization, batch effects, clustering choices, dimensionality reduction, trajectory inference,

pathway annotation, feature selection, and, in spatial datasets, segmentation and resolution constraints.[29-33] A musical mapping that sounds coherent does not necessarily imply biological causality, and a musically striking pattern may reflect preprocessing decisions rather than true biological structure.[16, 26, 28] For this reason, musification should be paired with conventional transcriptomic analysis, such as differential expression testing, clustering validation, pathway enrichment, co-expression analysis, trajectory inference, or spatial-domain analysis.[29-33] The most scientifically responsible approach is to treat musification as an exploratory, communicative, educational, and aesthetic layer that can reveal perceptual patterns, support education, and broaden access to complex data, but not as a substitute for statistical inference.

Overall, transcriptomic data have strong potential for biological music generation because they are dynamic, quantitative, high-dimensional, and deeply connected to cellular function. Their most distinctive musical affordance is not single-gene melody, but the possibility of converting coordinated expression programs into polyphonic, temporally evolving, and functionally annotated musical systems.



Proteomic and Structural-Biological Data: From Amino Acid Sequences to Folded Molecular Music

The hierarchical substrate of proteomics.

Proteomic and structural-biological data provide one of the richest substrates for biological musification because proteins can be represented across multiple levels of organization. Unlike genomic sequences, which are primarily linear, symbolic, and discrete, proteins are simultaneously sequential, chemical, structural, dynamic, quantitative, and interactive. At the single-protein level, a molecule may be treated as an amino-acid sequence, a distribution of physicochemical properties, a pattern of secondary-structure elements, a folded three-dimensional architecture, a conformational ensemble, a vibrational system, or a post-translationally modified entity. At the broader proteome scale, abundance, localization, pathway membership, modification state, and protein-protein interaction information could further expand the available musical mapping space. This hierarchy makes proteomic musification especially suitable for translating molecular organization into musical organization: pitch and rhythm may encode sequence or residue-level properties; timbre and register may reflect physicochemical character; harmony and texture may represent folding, secondary structure, or interaction patterns; dynamics and density may convey abundance or modification state; and spatialization or large-scale form may be used to represent three-dimensional architecture, conformational change, or systems-level relationships.[9, 10, 13, 19-21, 34-36]

Sequence-based protein music and symbolic translation.
Early protein musification often began from the amino-acid sequence itself. King and Colin's PM — Protein Music program and Dunn and Clark's Life Music project are foundational examples of this sequence-oriented approach, translating amino-acid order into musical structures and treating molecular sequence as audible pattern.[9, 10] In this sequence-oriented approach, amino-acid identity may be mapped to pitch, recurring sequence motifs may become melodic figures, and substitutions or insertions may produce variation. The strength of this method is its accessibility: proteins, like DNA, can be read as linear strings and converted into temporal musical sequences. Its limitation is that a simple residue-to-note mapping risks reducing proteins to symbolic chains while ignoring the chemical and folded properties that distinguish protein data from genomic data.

Physicochemical mapping and the folding metaphor.
Later approaches expanded protein musification beyond alphabetic translation by incorporating physicochemical and structural information. Bywater and Middleton explored whether melodies derived from evolutionary information, predicted secondary structure, flexibility, hydropathy, and solvent accessibility could support protein fold classification.[34] Martin, Meagher, and Barker developed parameter-mapping algorithms for protein sequences and multiple sequence alignments, using sound as a complementary channel for perceiving sequence features and alignment patterns.[13] These studies show that protein music can encode not only residue order, but also biologically interpretable properties such as hydrophobicity, charge, flexibility, conservation, secondary structure, and solvent exposure. In musical terms, this means that pitch alone is insufficient: register, rhythm, articulation, timbre, dynamics, and texture can all carry different layers of protein information.

Biophysical grounding and molecular vibrations.
A more physically grounded model is represented by Yu, Qin, Martin-Martinez, and Buehler's self-consistent sonification method.[20] Instead of assigning amino acids to arbitrary notes, they calculated the normal-mode vibrations of the 20 natural amino

acids and transposed these molecular vibrations into the audible range. In this framework, each amino acid is associated with a sound derived from its own vibrational properties, while secondary-structure information can further control note duration and volume.[20] This method is important because it connects sonic identity to molecular mechanics rather than purely symbolic convention. It also suggests a broader design principle for proteomic musification: the more dimensions of protein biology are incorporated, the more the resulting music can reflect molecular organization rather than merely decorate a sequence.

Spectroscopic translation via NMR.
Spectroscopy-informed approaches further broaden protein musification by grounding musical parameters in experimentally derived molecular signals. Monajjemi's 13C-NMR sonification of human insulin represents an earlier attempt to connect amino-acid sequence translation with NMR-derived molecular information, while more recent NMR-informed frameworks have explored how spectral and residue-level features may contribute to protein-sequence and function-oriented musical representations.[25, 36] NMR data, for example, can provide information related to residue environment, conformational state, and protein dynamics, allowing sonic mappings to move beyond primary sequence alone. In such designs, spectral peaks, chemical-shift patterns, or residue-level structural annotations may be translated into pitch, resonance, timbre, or rhythmic emphasis. This approach is valuable because it links musical features to experimentally observed molecular behavior rather than relying solely on predicted sequence properties.[25]

Intersections with AI-driven protein design.
Yu and colleagues further linked sonified amino-acid representations with recurrent neural networks for de novo protein sequence generation, suggesting that sound can function not only as an auditory output but also as an intermediate representation for protein design.[20] However, such AI-assisted workflows should be interpreted as exploratory complements to structural prediction, molecular simulation, biochemical characterization, and experimental validation.

Protein musification in viral structural biology.
The SARS-CoV-2 spike protein provides a prominent example of how protein musification can intersect with structural biology, molecular mechanics, and public-facing science communication. Kolel-Veetil and colleagues discussed SARS-CoV-2 infection through the combined lens of music and mechanics, using the spike protein as a structurally and functionally iconic biomolecular system.[21] This example illustrates that protein music can be used not only to translate amino-acid sequences, but also to frame higher-order questions about protein architecture, conformational behavior, and mechanical function. At the same time, such representations should remain coupled to structural, mechanical, and biochemical evidence, rather than being interpreted as independent proof of viral function or pathogenicity.

Proteomic scale and evolutionary variations.

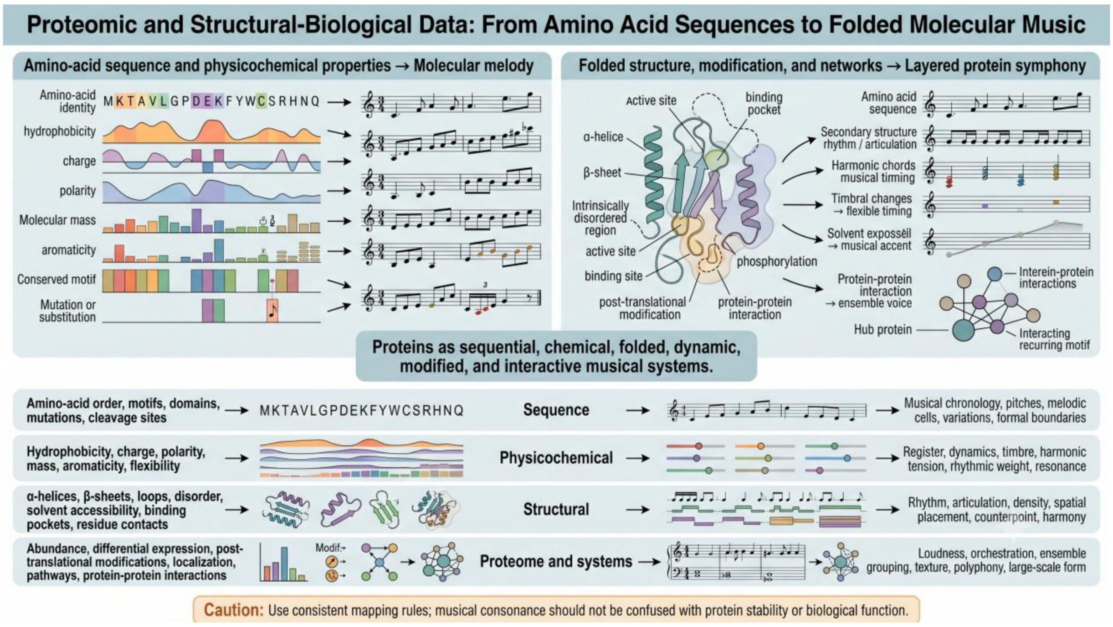
At larger scales, protein musification can compare homologous proteins, isoforms, mutations, or protein families rather than representing a single sequence in isolation.[13, 34] Conserved residues may become recurring musical anchors, while substitutions, insertions, deletions, or domain rearrangements can generate variation, modulation, or structural disruption. Together, these approaches suggest that proteomic musification exists on a continuum between sequence-based composition, analytical sonification, and molecularly grounded sound design. Dunn and Clark demonstrate how amino-acid order can generate musical material; Bywater and Middleton show how fold-related features can be made audible; Martin and colleagues emphasize parameter mapping for sequence and alignment analysis; and Yu and colleagues connect protein sound to molecular vibration and AI-assisted protein design.[9, 13, 20, 34] Compared with genomic musification, however, proteomic musification offers a broader and more layered creative field because proteins carry many more biologically meaningful variables at once. For creators and researchers, the key challenge is therefore not finding enough attributes to map, but choosing a small number of interpretable mappings that preserve both musical clarity and biological traceability.

Synthesizing these metabolomics and network-analysis perspectives, metabolomic musification can be organized across four functional layers. First, at the sequence layer, amino-acid order provides musical chronology. Residue identity, repeated motifs, conserved domains, mutations, or cleavage sites can become pitches, melodic cells, variations, or formal boundaries. Second, at the physicochemical layer, amino-acid properties such as hydrophobicity, charge, polarity, molecular mass, aromaticity, or flexibility can control register, dynamics, timbre, harmonic tension, or rhythmic weight. Third, at the structural layer, α-helices, β-sheets, loops, intrinsically disordered regions, solvent accessibility, binding pockets, and residue contacts can shape rhythm, articulation, density, spatial placement, or counterpoint. Fourth, at the proteome or systems layer, abundance, differential expression, post-translational modifications, subcellular localization, pathway membership, and protein-protein interactions can determine orchestration, loudness, ensemble grouping, texture, and large-scale musical form.[9, 13, 20, 34, 35]

For artists and composers, the most accessible starting point is not to sonify an entire proteome immediately, but to choose a biologically meaningful protein, protein family, mutation comparison, pathway, or small proteomic dataset. A practical beginner strategy is to use the amino-acid sequence as the timeline, map one physicochemical property such as hydrophobicity or charge to register or timbre, use secondary structure to shape rhythm or articulation, and use functional annotations such as active sites, mutations, or post-translational modifications as musical accents. For researchers, the same logic can be made more analytical by keeping mapping rules fixed across samples or conditions: for example, abundance may control loudness, phosphorylation may introduce timbral change, and protein-protein interactions may determine which instrumental voices sound together. In this way, proteomic musification can function both as molecularly inspired composition and as exploratory auditory display. Its distinctive advantage is that the resulting music can remain traceable not only to sequence, but also to folding, chemistry, modification, abundance, and biological function.

Methodological caveats and analytical limitations.
Despite this richness, proteomic musification also carries a high risk of overmapping. Because proteins contain many available features, assigning too many variables simultaneously can produce music that is biologically traceable in principle but perceptually opaque in practice. Structural data may also vary in confidence, especially when predicted models, intrinsically disordered regions, or conformational ensembles are involved. Similarly, abundance and post-translational modification data depend on experimental platform, normalization, coverage, and detection sensitivity. Therefore, proteomic musification should use a limited number of biologically justified mappings, report data sources and preprocessing choices clearly, and avoid equating musical consonance, dissonance, or beauty with protein stability, function, or evolutionary fitness.[1, 2]

A hierarchical framework for structural musification.
The evolution of protein musification reveals a stark contrast between abstract symbolic translation (which ignores 3D reality) and pure quantum-vibrational sonification (which can obscure the linear sequence narrative). Bridging this gap requires a hierarchical, multi-dimensional mapping framework that respects the physical realities of protein folding. For practitioners, a robust compromise involves distributing biological features across distinct musical dimensions: the primary amino-acid sequence dictates the melodic timeline; physicochemical properties (e.g., hydrophobicity) govern vertical register and harmonic tension; secondary structures (α-helices vs. β-sheets) drive rhythmic articulation; and 3D residue contacts define spatial audio placement. This layered approach ensures that the resulting composition is not merely "inspired" by a protein, but can serve as a structurally informed acoustic representation of molecular folding.[9, 13, 20, 25, 34, 35]



Metabolomic Data: From Molecular Abundance to Biochemical Network Music

Metabolic networks as the shifting score.

Metabolomic data occupy a distinctive position within biological musification because they are closer to phenotype than genomic or proteomic sequences, yet less naturally linear than either DNA or amino-acid chains. Whereas genomic data can be read as symbolic strings and proteomic data can be organized through sequence, structure, and physicochemical hierarchy, metabolomic data are best understood as dynamic biochemical states. A metabolome reflects the integrated outcome of enzyme activity, pathway regulation, environmental exposure, nutrition, disease status, and cellular context. For this reason, metabolomic musification should not be limited to assigning individual metabolites to isolated notes. Instead, it is better framed as the musical translation of biochemical relationships: metabolite abundance, pathway membership, reaction topology, disease-associated perturbation, network connectivity, and annotation confidence can all become compositional variables. This creates a conceptual shift from “sequence as melody” to “network as score,” in which metabolic graphs, rather than linear molecular strings, provide the structural basis for musical form.[18, 37-39]

Abundance mapping and the flux limitation.

A common starting point for metabolomic musification is metabolite abundance. Relative concentration, fold change, statistical significance, or condition-specific variation can be mapped to pitch, loudness, duration, register, density, or timbre. For example, increased abundance may be represented by higher register or greater intensity, while depleted metabolites may appear as lower, quieter, or harmonically altered events. This type of mapping is intuitive and can make differential metabolic states audible. However, abundance-based sonification has an important biological limitation: metabolite concentration does not directly equal metabolic flux. A metabolite may accumulate because it is produced more rapidly, consumed more slowly, transported differently, or regulated through pathway bottlenecks. Therefore, music derived only from abundance values may represent biochemical state, but not necessarily reaction direction, pathway activity, or causal mechanism.[39, 40]

Isotope tracing and dynamic biochemical motion.
Isotope tracing provides a more dynamic basis for metabolomic musification because it follows labeled substrates through metabolic pathways and can reveal turnover, pathway usage, and metabolic flow. In a musical setting, isotope-labeling patterns could be mapped to rhythmic transformation, temporal delay, melodic direction, or evolving harmonic progression. For instance, the gradual incorporation of labeled carbon into downstream metabolites could become a progressive change in timbre or motif density, while competing pathway routes could be represented by diverging instrumental voices. This distinction between abundance and flux is central to rigorous metabolomic musification. Abundance-based mappings are suitable for representing metabolic states, whereas flux-informed mappings are more appropriate for representing biochemical movement, transformation, and directionality.[40]

Untargeted feature mapping and network integration.
Untargeted metabolomics creates both an opportunity and a challenge for musical translation. Its strength is that it captures a broad range of molecular features without restricting the analysis to a predefined metabolite list. Its difficulty is that many detected features remain incompletely identified or ambiguously annotated. Directly mapping every detected peak or feature to a musical note can therefore produce an acoustically rich but biologically uncertain result. Network integration offers a more interpretable strategy. Pirhaji and colleagues showed that untargeted metabolomic features can be integrated into molecular networks to reveal disease-associated pathways, even when many individual features are not fully identified.26 This is especially relevant for musification because it shifts the musical object from a list of metabolite names to a structured biochemical system. Nodes, edges, subnetworks, pathway proximity, and disease-associated modules can each be translated into musical events, interactions, textures, or formal sections.[18, 38, 39]

Systems biology as a conceptual bridge.
Systems biology provides the conceptual bridge between metabolomic data and network-based music. Pathway enrichment, metabolic modeling, correlation networks, reaction topology, and network centrality can all help organize metabolomic signals into biologically meaningful structures. Rosato and colleagues emphasize that metabolomic data should be interpreted through systems-level approaches rather than through isolated metabolite changes alone.[39] In musical terms, this means that a metabolite does not need to function as a single note in

isolation; it may instead behave as part of a chord, motif, instrumental group, or pathway-specific texture. Highly connected metabolites may recur as rhythmic anchors, central pathway nodes may become tonal centers, and perturbed subnetworks may generate contrasting musical sections. Slenter's thesis makes this idea explicit by treating biochemical graphs as sheet music, positioning metabolic networks as compositional structures rather than merely as data sources.[18] In this framework, musification becomes a way of rendering pathway architecture, biochemical interaction, and disease perturbation audible.

Encoding annotation uncertainty via acoustics.

A major methodological issue in metabolomics is that metabolite identification is not equally certain across all detected signals. Schymanski and colleagues proposed confidence levels for small-molecule identification by high-resolution mass spectrometry, distinguishing confirmed structures from probable structures, tentative candidates, and unknown compounds.[41] This principle is crucial for metabolomic musification because uncertain annotations should not be presented as if they were confirmed biochemical entities. Instead, confidence itself can be encoded musically. Confirmed metabolites may be represented by stable timbres, clear attacks, central register, or prominent motifs; tentative annotations may use softer dynamics, unstable timbre, ornamentation, or peripheral spatial placement; unknown features may appear as background texture, noise-like material, or unresolved harmonic color. In this way, the music can preserve uncertainty rather than concealing it. Annotation confidence should therefore be treated not as a technical detail outside the composition, but as one of the core parameters of metabolomic sound design.[41, 42]

Standardizing the computational pipeline.

Because metabolomic musification depends heavily on preprocessing, normalization, annotation, and mapping decisions, transparent reporting is essential. The Metabolomics Standards Initiative proposed minimum reporting standards for chemical analysis, emphasizing the need to document experimental and analytical procedures.[42] For musical translation, similar transparency is needed. A metabolomic musification study should report the biological sample or dataset, analytical platform, preprocessing pipeline, normalization method, statistical threshold, metabolite annotation level, pathway or network database, network construction method, and the exact rules used to map biochemical variables into musical parameters. Without these details, a metabolic composition may be aesthetically compelling but scientifically difficult to interpret or reproduce. In this sense, a metabolomic score is not only a musical object but also a methodological record of data selection, transformation, and interpretation.[1, 2, 41, 42]

Auralization paradigms and predictive interfaces.

Compared with DNA and protein sonification, metabolomic musification remains relatively underdeveloped, but recent work suggests an emerging computational direction. NetAurHPD, for example, uses network auralization in the context of hyperlink prediction to identify metabolic pathways from metabolomics data.[37] This indicates that metabolomic sound may move beyond artistic representation toward analytical and predictive auditory interfaces. When combined with graph models, machine learning, pathway databases, and isotope-informed flux analysis, metabolomic musification could support new ways of exploring biochemical state changes, disease-associated networks, and hidden pathway relationships. However, such approaches should be interpreted carefully: auditory pattern recognition can complement visualization and statistical analysis, but it should not replace biochemical validation or causal inference.[37-40]

A four-layered functional framework.

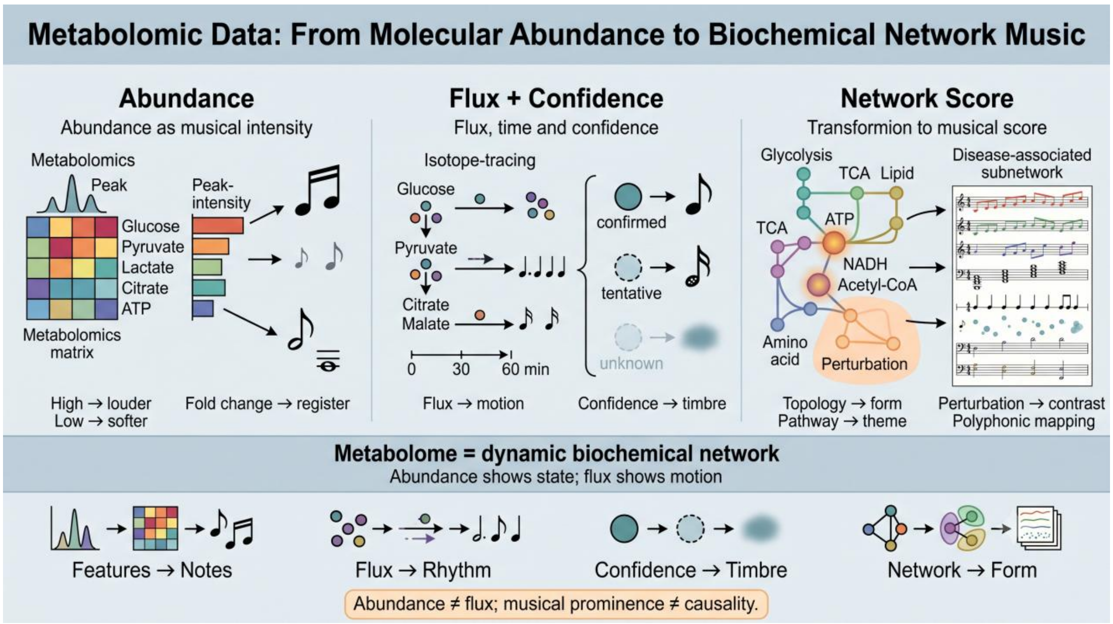
Synthesizing these metabolomics and network-analysis perspectives, metabolomic musification can be organized across four functional layers rather than a single linear mapping. First, individual metabolites or mass-spectral features provide basic sound events; their abundance, fold change, or statistical significance may control pitch, loudness, duration, or intensity. Second, metabolite identification confidence can shape timbre, clarity, register, spatial position, or prominence, ensuring that confirmed and uncertain features are not musically equivalent. Third, pathway membership, enrichment, reaction direction, and isotope-labeling dynamics can determine motifs, rhythm, harmonic progression, or temporal development. Fourth, graph topology, centrality, modularity, pathway connectivity, and disease-associated subnetworks can shape orchestration, counterpoint, recurrence, density, and large-scale form.[18, 37-42]

Methodological caveats and risks of overmapping.

Despite its promise, metabolomic musification carries several risks. The first is overmapping: metabolomic datasets contain abundance values, feature intensities, statistical scores, annotations, pathway memberships, and network properties, but assigning all of these simultaneously can produce an auditory result that is technically traceable yet perceptually opaque. The second is overinterpretation: a musically prominent event may reflect a strong fold change, high network centrality, or confident annotation, but it should not automatically be interpreted as biological causality or clinical importance. The third is uncertainty: unknown features, ambiguous annotations, missing pathway information, and platform-specific biases are common in metabolomics. Therefore, metabolomic musification should use a limited number of biologically justified mappings, distinguish abundance from flux, encode annotation confidence, report preprocessing and mapping rules, and avoid treating musical consonance, dissonance, or beauty as direct evidence of biochemical function.[1, 2, 40-42]

The holistic network-to-score synthesis.

The most rigorous direction for metabolomic musification is therefore not a simple metabolite-to-note dictionary, but a network-based framework that treats biochemical organization as musical organization. In such a framework, metabolite abundance provides local intensity, isotope tracing provides temporal movement, annotation confidence provides sonic certainty, pathway membership provides thematic grouping, and network topology provides large-scale form. This layered approach allows metabolic music to remain both aesthetically organized and biologically traceable. More importantly, it respects the distinctive nature of metabolomic data: the metabolome is not a linear text to be read note by note, but a dynamic biochemical network whose meaning emerges from relationships, flows, perturbations, and uncertainty.[18, 37-42]



Genetic Variation and Association Studies: From Statistical Peaks to Polygenic Soundscapes

Population-level statistics rather than molecular sequence.

Genetic variation and association studies introduce a distinct statistical layer into biomolecular musification. Unlike genomic, proteomic, or metabolomic data derived from individual molecular entities, genome-wide association studies (GWAS) and epigenome-wide association studies (EWAS) are population-level association landscapes. GWAS typically relate genetic variants, most often single-nucleotide polymorphisms, to phenotypes or disease traits across large cohorts, whereas EWAS examine associations between phenotypes or exposures and epigenetic marks, most commonly DNA methylation at CpG sites.[43, 44] Their central musical substrate is therefore not a molecular sequence, folded structure, or biochemical pathway, but a high-dimensional matrix of genomic coordinates, association statistics, effect sizes, allele frequencies, methylation differences, linkage disequilibrium structure, and phenotype-level uncertainty.[43-47]

Manhattan plots as auditory timelines.

The most natural starting point for GWAS or EWAS musification is the Manhattan plot. In a conventional Manhattan plot, genomic coordinates are arranged along the horizontal axis and association strength is represented by $-\log_{10} P$ values on the vertical axis.[43, 48] This structure maps naturally onto musical time and salience. Chromosomal position can define the temporal order of the piece, while $-\log_{10} P$ values may control pitch height, loudness, brightness, register, rhythmic density, or musical prominence. Variants or CpG sites below the significance threshold may form a low-density background texture, whereas loci or methylation sites exceeding genome-wide or epigenome-wide significance may be rendered as accented tones, registral peaks, percussion markers, or short recurring motifs. Effect size and direction can further modulate dynamics, stereo position, melodic contour, or instrumental color. In this framework, sonification functions as an auditory genome browser: it allows statistically prominent regions to become perceptually salient during a genome-wide scan.

LD blocks and polygenic texture.

A key challenge is that association peaks should not be interpreted as isolated causal notes. In GWAS, nearby variants are often correlated through linkage disequilibrium, meaning that an associated SNP may tag a broader genomic region rather than directly cause the phenotype.[45, 48] For this reason, LD blocks may be more appropriately represented as local harmonic fields, sustained clusters, or textured regions rather than as single, isolated notes. Likewise, many common diseases and behavioral traits have highly polygenic architectures, in which large numbers of variants each contribute small effects.[46, 47] A polygenic trait may therefore be better translated into distributed rhythmic density, cumulative orchestration, or slowly evolving harmonic tension than into a few dramatic alarm tones. Polygenic scores could be represented as aggregate musical intensity or ensemble density, but such mappings should be interpreted as summaries of statistical risk or trait association, not as deterministic predictions of individual biological destiny.[46, 47]

EWAS and environmentally responsive regulatory soundscapes.

EWAS adds another dimension because epigenetic marks are dynamic, tissue-specific, environmentally responsive, and potentially confounded by cell composition, exposure history, disease state, and technical variation.[44, 49] In musical terms, EWAS data can be treated as coordinate-based regulatory association landscapes. CpG sites or differentially methylated regions may define musical events along genomic time, while methylation difference, test statistic, or $-\log_{10} P$ value can control pitch, loudness, timbre, or rhythmic emphasis. However, because DNA methylation is strongly context-dependent, an EWAS-derived musical event should not be interpreted as a stable inherited genetic signal. Instead, it may be better represented through timbral instability, modulation, or conditional texture, emphasizing that epigenomic associations often reflect dynamic biological and environmental states rather than fixed sequence variation.[44, 49]

Dissonance as statistical salience, not biological causality.
Dissonance can be a useful perceptual cue in association-data musification, but it requires careful interpretation. A sharp dissonant interval, noisy texture, or sudden rhythmic disruption may effectively draw attention to a significant locus, highly differentiated methylation site, or unusual association peak. However, dissonance should represent statistical salience, uncertainty, or deviation from a reference pattern, not confirmed pathogenicity or molecular damage. Many GWAS loci require fine-mapping, functional annotation, experimental validation, and biological interpretation before causal claims can be made.[45, 48] Similarly, EWAS associations may reflect exposure, disease consequence, cell-type composition, or confounding rather than direct regulatory causality.[44, 49] Therefore, musical prominence should be understood as a guide for attention, not as evidence of mechanism.

Musical traits as genetic phenotypes and AI-assisted future directions.
Recent work on the genetics of musicality also provides an inverse perspective on the relationship between music and genomics. A large-scale GWAS of beat synchronization in more than 600,000 individuals reported a highly polygenic architecture, 69 genome-wide significant loci, SNP-based heritability estimates of approximately 13–16%, enrichment in brain-expressed genes and brain-specific regulatory elements, and genetic correlations with motor function, breathing-related traits, processing speed, and chronotype.[50] Broader behavioral-genetic work further suggests that music-related traits are not unitary phenotypes: music engagement shows genetic relationships with mental-health-related traits, while twin modelling of music enjoyment indicates partly distinct genetic pathways underlying different dimensions of musical reward and engagement.[51, 52] These studies do not directly sonify GWAS or twin-model data, but they demonstrate that music-related behaviors themselves can be analyzed as complex, partially heritable, and genetically heterogeneous phenotypes. In parallel, studies of pre-trained Transformer models suggest that sequence-learning architectures can transfer across text, amino-acid, DNA, and symbolic music datasets.[23] These findings point toward future generative sonification systems in which statistical genomic signals, behavioral-genetic models, sequence representations, and musical structures are integrated. However, such applications remain exploratory and should not be presented as validated diagnostic tools.

Practical framework for association-data musification.
A practical framework for GWAS and EWAS musification can be organized across four layers. First, the coordinate layer maps chromosome, genomic position, or CpG location to musical time. Second, the association layer maps $-\log_{10} P$, test statistics, effect size, methylation difference, or confidence interval width to pitch, loudness, brightness, rhythmic density, or accent. Third, the regional layer maps LD blocks, fine-mapped credible sets, differentially methylated regions, genes, enhancers, or regulatory annotations to harmonic clusters, motifs, timbral regions, or instrumental groups. Fourth, the trait-level layer maps polygenic scores, genetic correlations,

pathway enrichment, tissue enrichment, or cross-trait associations to large-scale orchestration, form, and polyphonic texture. This layered approach allows GWAS and EWAS data to become audible while preserving their central limitation: they are statistical association landscapes, not direct musical proof of causality.

For this reason, association-data musification should distinguish among GWAS summary statistics, mutational-signature models, twin-model estimates, and polygenic scores, because these data types differ in statistical meaning and should not be collapsed into a single "genetic music" representation.[27, 50-52]

Cross-Modality Design Principles for Biomolecular Musification

Across genomic, epigenomic, transcriptomic, proteomic, metabolomic, and association-study data, a central lesson emerges: biologically meaningful musification should not begin with a universal note dictionary, but with the dominant structure of the data itself. Genomic sequences are discrete and linear, making codons, reading frames, motifs, and repeats natural sources of musical chronology. Epigenomic data are coordinate-based regulatory landscapes, making peaks, chromatin states, methylation patterns, and comparative regulatory transitions suitable for texture, density, timbre, and form. Transcriptomic data are high-dimensional expression matrices, making time points, gene clusters, pathways, cell states, and trajectories more appropriate musical units than isolated genes. Proteomic data are hierarchical molecular systems, allowing sequence, physicochemical properties, secondary structure, tertiary contacts, post-translational modifications, and interaction networks to occupy different musical layers. Metabolomic data are dynamic biochemical networks, requiring abundance, flux, annotation confidence, pathway membership, and network topology to be treated as interacting musical dimensions. Association-study data are statistical genomic landscapes, making genomic coordinates, association strength, LD structure, effect size, and uncertainty more appropriate musical units than individual variants alone.

The strongest mapping strategies are therefore modality-specific and hierarchical. A useful principle is that the most biologically important organizational axis should control the most musically important organizational axis. If genomic order is central, it should determine musical time. If expression dynamics are central, biological time points should shape musical development. If protein folding is central, structural hierarchy should guide rhythm, articulation, spatialization, and harmony. If metabolic network topology is central, graph structure should shape recurrence, orchestration, and large-scale form. This principle is consistent with perceptual approaches to auditory display, which emphasize that sonification should be designed around listener discrimination, task relevance, and interpretability rather than around maximal data-to-sound transfer.[4] This approach prevents musification from collapsing into arbitrary sonification, while still allowing musical creativity within biologically defined constraints.

A second design principle is perceptual parsimony. Auditory display research has long emphasized that sonification succeeds only when listeners can perceive, segregate, and interpret the encoded variables.[4] Multi-omics datasets contain many potential variables, but mapping too many features simultaneously can produce an acoustic output that is biologically traceable in theory yet perceptually unintelligible in practice. Effective musification usually requires selecting a limited number of biologically justified variables and assigning each to a perceptually distinguishable musical dimension. Pitch, loudness, timbre, rhythm, articulation, register, spatialization, and density should not be treated as interchangeable containers; each parameter has different perceptual strengths and limitations. For example, pitch is effective for ordered or graded relationships, rhythm for temporal or periodic processes, timbre for categorical identity or confidence, and spatialization for anatomical, structural, or network position.

A third principle is consistency and traceability. The same biological feature should be mapped in the same way across samples, conditions, or time points unless a change in mapping is explicitly justified. This is especially important in comparative musification, where the listener is asked to hear differences between healthy and diseased tissue, treated and untreated cells, wild-type and mutant proteins, or annotated and uncertain metabolites. Without consistent mapping rules, musical contrast may reflect compositional choice rather than biological difference. For this reason, musification studies should report datasets, preprocessing steps, normalization methods, feature-selection criteria, mapping rules, software tools, and, where possible, source code or reproducible audio-generation pipelines.

A fourth principle is comparative design. Absolute sonification of a single molecular object may be aesthetically compelling, but comparison often carries greater biological meaning. In genomic musification, mutations or motifs become informative when heard against a reference sequence. In epigenomic musification, regulatory activation or repression is most meaningful when contrasted across cell states or disease conditions. In transcriptomics, expression trajectories become interpretable when clusters, treatments, lineages, or time points are compared. In proteomics, substitutions, domains, post-translational modifications, and interaction changes are clearer when mapped relative to a homolog, structural model, or functional state. In metabolomics, abundance changes, pathway perturbations, and flux shifts are most meaningful when heard against a reference metabolic state.

Finally, musification must preserve uncertainty. Biological datasets are shaped by sequencing depth, normalization, batch effects, annotation confidence, structural prediction quality, missing values, instrument sensitivity, and statistical thresholds.

These uncertainties should not be hidden by aesthetically polished musical output. Instead, confidence and uncertainty can themselves become musical parameters: stable timbres for confirmed annotations, blurred textures for uncertain features, quieter layers for low-confidence signals, or spatially peripheral placement for tentative assignments. Such strategies allow music to communicate not only biological pattern but also epistemic status.

Taken together, these principles position biomolecular musification as a complementary representational practice rather than a replacement for statistical analysis, visualization, or experimental validation. Musical consonance should not be equated with molecular stability; dissonance should not be equated with pathogenicity; loudness should not be equated with biological importance; and aesthetic beauty should not be equated with functional truth. The value of musification lies in its ability to create an additional perceptual and cultural interface for complex biological data—one that can emphasize temporality, pattern, recurrence, contrast, and layered organization in ways that visual representations may not. It can also create productive spaces for collaboration among scientists, musicians, educators, and sound artists, provided that artistic translation remains clearly distinguished from analytical validation.[3]

Conclusion: Toward a Rigorous Biology of Musical Representation

Biomolecular musification has evolved from a symbolic curiosity into a diverse representational practice spanning genomic sequences, epigenomic landscapes, transcriptomic dynamics, protein structures, and metabolic networks. Early DNA-to-note translations demonstrated that biological sequences could be experienced as ordered temporal material, but modern approaches increasingly recognize that each biological data type has its own syntax, hierarchy, uncertainty, and scale. A genome is not an expression matrix; a chromatin track is not a folded protein; a metabolome is not a linear string. Consequently, the future of the field depends less on inventing universal musical alphabets and more on designing modality-specific mapping frameworks that respect the structure of the underlying data.

Across the examples reviewed here, the most compelling systems are those that preserve biological organization while exploiting the perceptual strengths of sound. Genomic musification is most effective when it incorporates codons, reading frames, motifs, and sequence recurrence. Epigenomic musification becomes meaningful when regulatory peaks, methylation states, and chromatin marks are treated as dynamic genomic landscapes. Transcriptomic musification gains power when gene expression is organized into clusters, trajectories, cell states, pathways, and functional backgrounds. Proteomic musification is strongest when amino-acid order is integrated with physicochemical properties, folding, molecular vibrations, post-translational modifications, and protein interactions. Metabolomic musification becomes most rigorous when it moves beyond metabolite-to-note translation and instead represents abundance, flux, pathway membership, annotation confidence, and biochemical network topology.

At the same time, the field must remain methodologically cautious. Sound can make data perceptually vivid, but perceptual vividness is not equivalent to biological explanation. Musical consonance does not demonstrate molecular stability, dissonance does not prove pathogenicity, and aesthetic coherence does not establish functional relevance. Musification should therefore be treated as a complementary layer of exploration, communication, accessibility, and artistic interpretation rather than as a substitute for statistical inference, molecular modeling, or experimental validation. For scientific use, future studies should report mapping rules transparently, provide reproducible code and datasets where possible, evaluate perceptual outcomes through listening studies, and compare auditory representations with established analytical methods.

The next stage of biomolecular musification will likely be shaped by three converging developments. First, multi-omics integration will require musical systems capable of representing several biological layers simultaneously, such as chromatin state, gene expression, protein abundance, and metabolic flux. Second, artificial intelligence and representation learning may enable more adaptive mappings, generative biological compositions, and bidirectional workflows in which sound becomes both an output for human listening and an intermediate feature space for computational models. Third, immersive and interactive audio technologies may allow researchers, students, artists, and visually impaired users to navigate molecular data through spatialized, manipulable sound environments.

Ultimately, the promise of biomolecular musification lies not in claiming that biology is literally music, nor in replacing visual analytics with sound. Its value lies in expanding the sensory vocabulary through which biological complexity can be encountered, while remaining grounded in perceptual principles of auditory display and in transparent collaboration between scientific and musical communities.[3, 4] Music can render recurrence, hierarchy, tension, transition, uncertainty, and networked interaction audible. When designed with biological rigor and musical sensitivity, biomolecular musification can serve as a bridge between molecular data and human perception, between scientific analysis and artistic creation, and between the hidden structures of life and the temporal intelligence of listening.

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